

PROJECT

SENSIFAKE

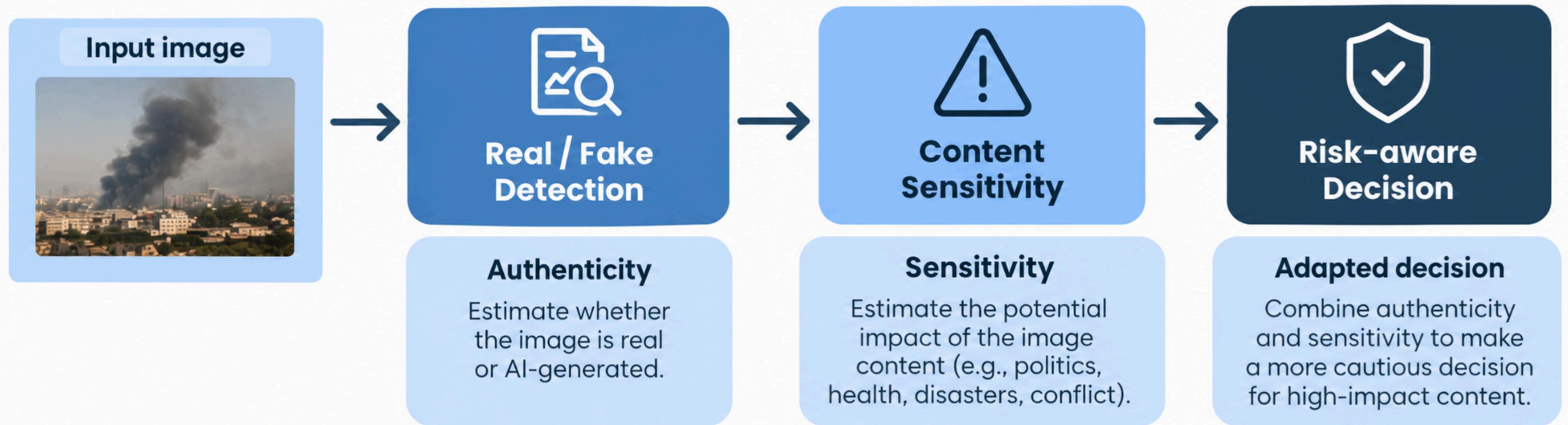
Sensitivity-aware detection of AI-generated images

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Project Goals

Not all classification errors have the same impact.

SensiFake aims to protect any individual from false accusations and to avoid the spreading of fabricated news by recognising the sensibility of the content of the image and consequently adapting its evaluation parameters.



SensiFake makes fake-image detection more cautious
when the consequences of an error are potentially higher.

Source Datasets

Approximately 1,500 images selected from each source

RRDataset — Real World Robustness

Images from seven real-world scenarios, including conflict, disasters, health, politics and everyday life.

OpenFake

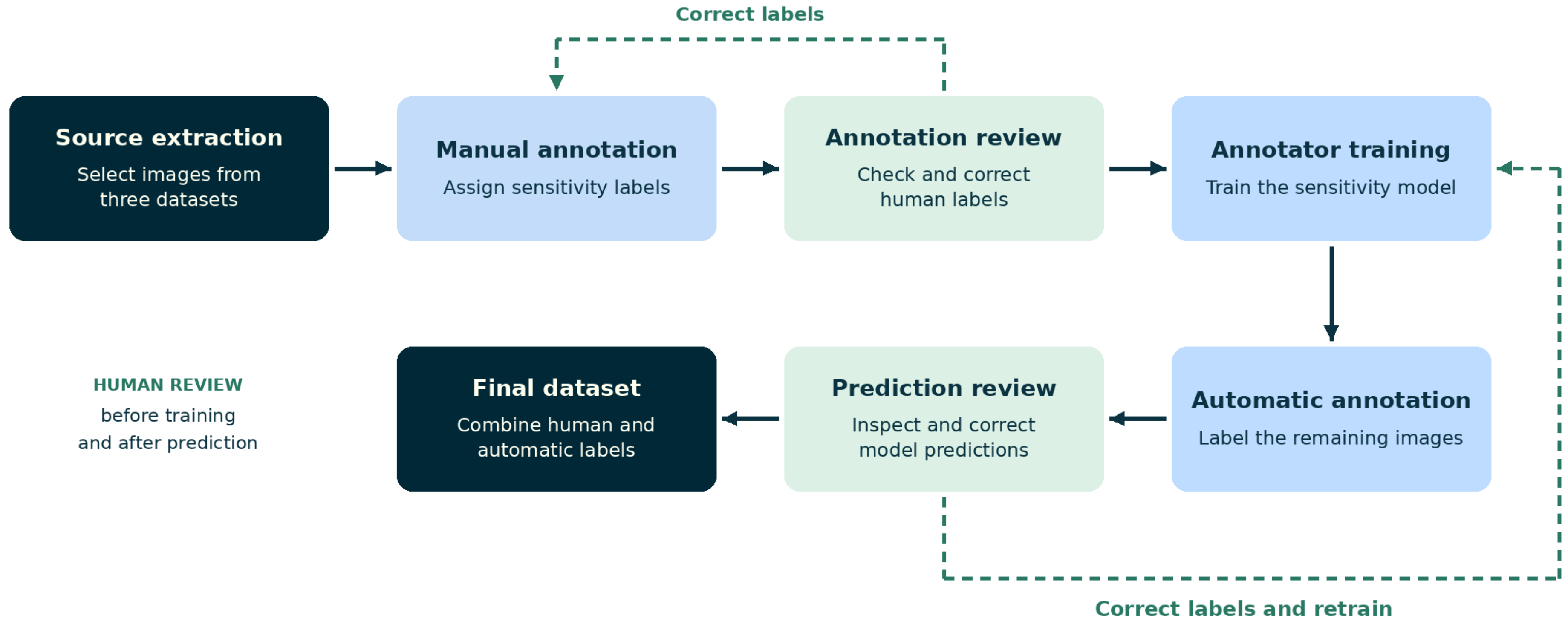
Dataset with a focus on politically and socially salient content

SID Set — Social media Image Detection

A social-media-oriented dataset of authentic, AI-generated and manipulated images.



Dataset Construction



Sensitivity Criteria

We assess content sensitivity through three criteria: public relevance, harm and vulnerability.

Public Relevance

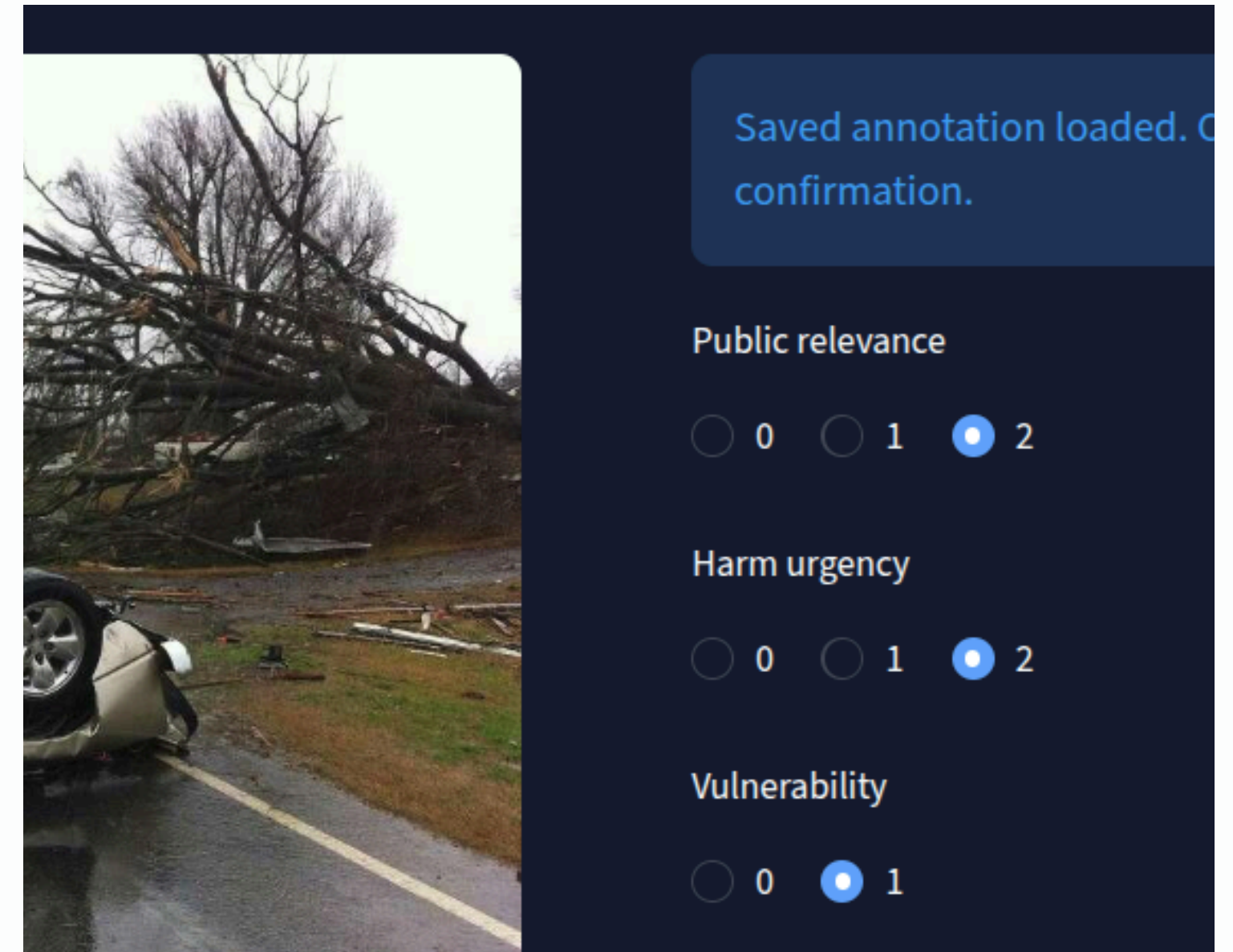
Potential impact on public debate, beliefs or decisions.

Harm urgency

Severity of the harm depicted in the image.

Vulnerability

Presence of people requiring additional protection.



Sensitivity Level

We assess content sensitivity through three criteria: public relevance, harm and vulnerability.

Low

No observable public impact or urgent harm.

Medium

Moderate public relevance or harm.

High

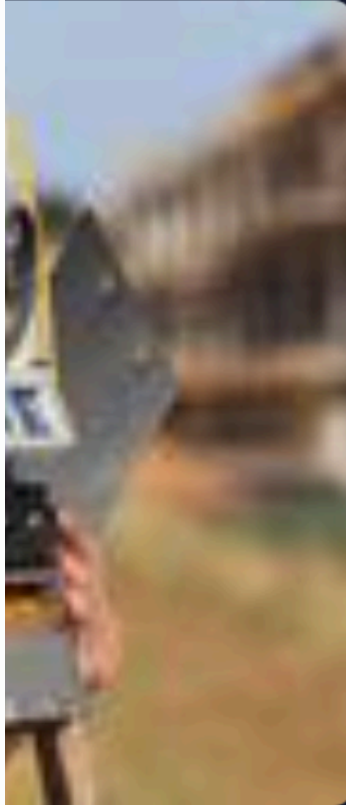
High public relevance, urgent harm, or vulnerability.

Queue position

95 / 600

Blind ID

SF-0095



Public relevance

☐ 0 ☒ 1 ☐ 2

Harm urgency

☒ 0 ☐ 1 ☐ 2

Vulnerability

☐ 0 ☒ 1

Sensitivity score

2 / 5

Sensitivity level

Medium

Optional observable rationale

Optionally describe visible cues (up to 280 characters).

Manual Annotation

One workspace for human annotation, label review and model prediction correction.

Workspace

☐ Annotate

☐ Review

☒ Model review

☐ Progress and exports

☐ Import batches

Batch

All batches ▾

SensiFake · Manual sensitivity annotation

Your name

Sarah

Resume or reserve an image

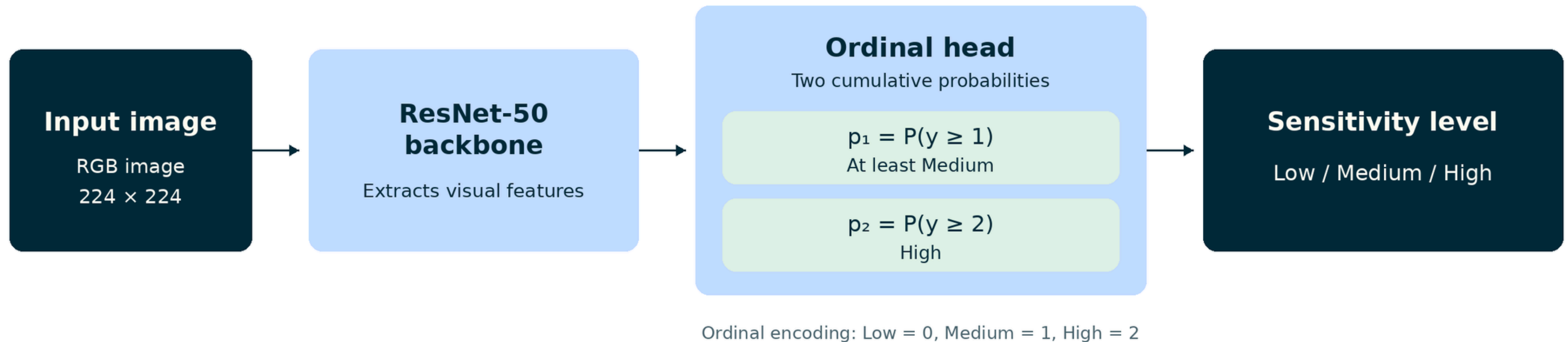
No pending model-review images are available. Check the batch and your assigned annotator name.

Each decision is saved as structured JSON, so it can be audited.

ANNOTATION - REVIEW - MODEL REVIEW

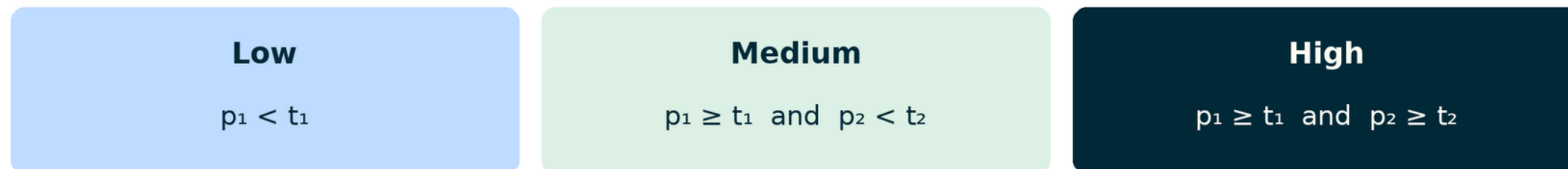
Automatic Annotation

An ordinal ResNet-50 model trained on human sensitivity labels



The ordinal formulation explicitly represents the ordering **Low < Medium < High**.

Decision rule



Final Annotated Dataset

4,499 unique images

Authenticity

2249 Real · 2250 Fake

Sensitivity predictions

1226 LOW · 3008 MEDIUM · 224 HIGH

Annotation record

1101 HUMAN · 3398 BY MODEL

Access the dataset



ACCURACY

75.7%

MACRO-F1

0.709

HIGH RECALL

66.7%

ORDINAL MAE

0.243

TEST RESULTS

Fixed test set · n = 140

Benchmark Setup

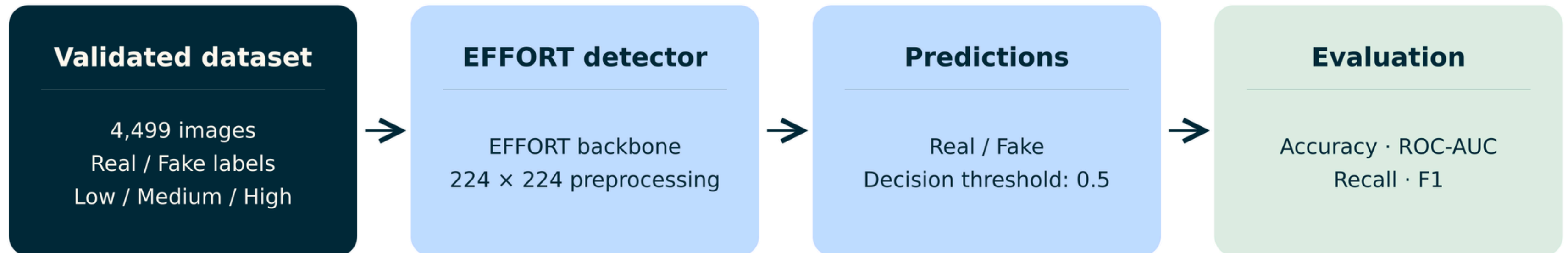
Evaluating EFFORT on 4,499 images across sensitivity levels.

To perform the benchmark the model used was Effort, a method integrated into **DeepfakeBench** for generalizable AI-generated image (AIGI) and deepfake detection.

The sensitivity metric was introduced by the following formula:

$$\frac{\sum_{i=1}^N w_i \cdot \mathbf{1}(\hat{y} = y_i)}{\sum_{i=1}^N w_i}$$

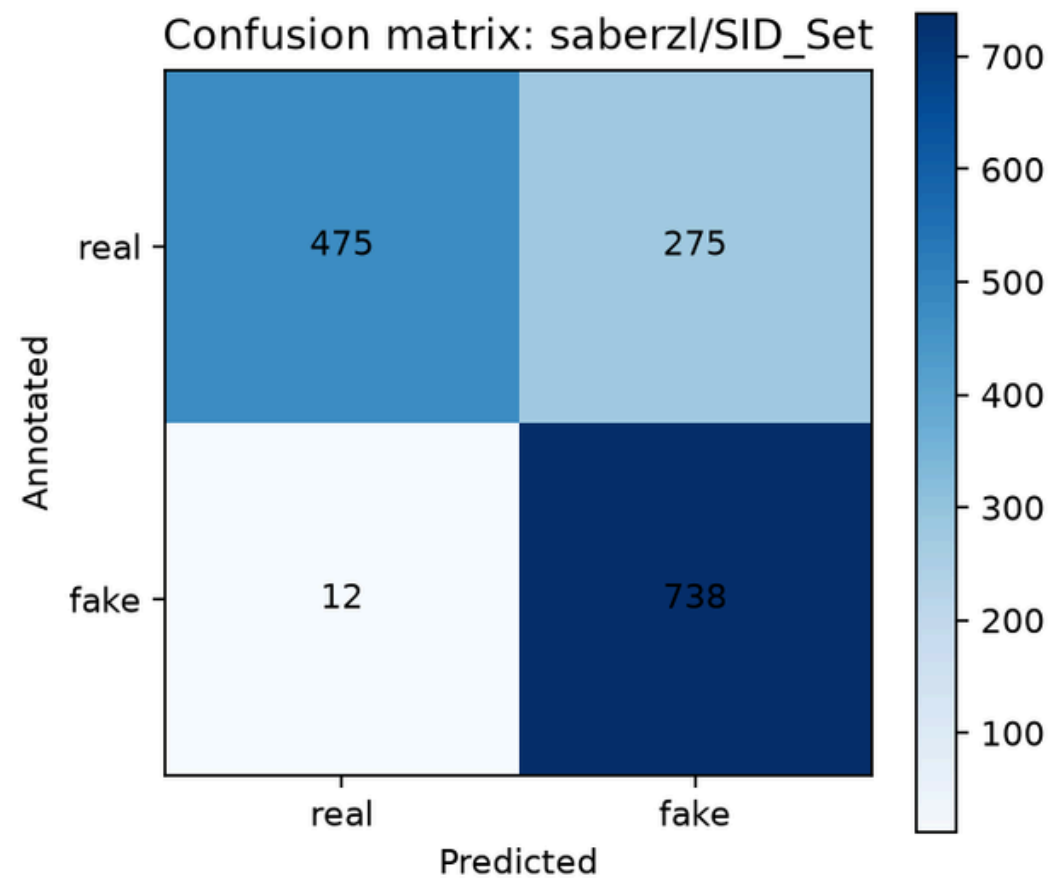
This way an error on a higher sensitivity image is worth more.



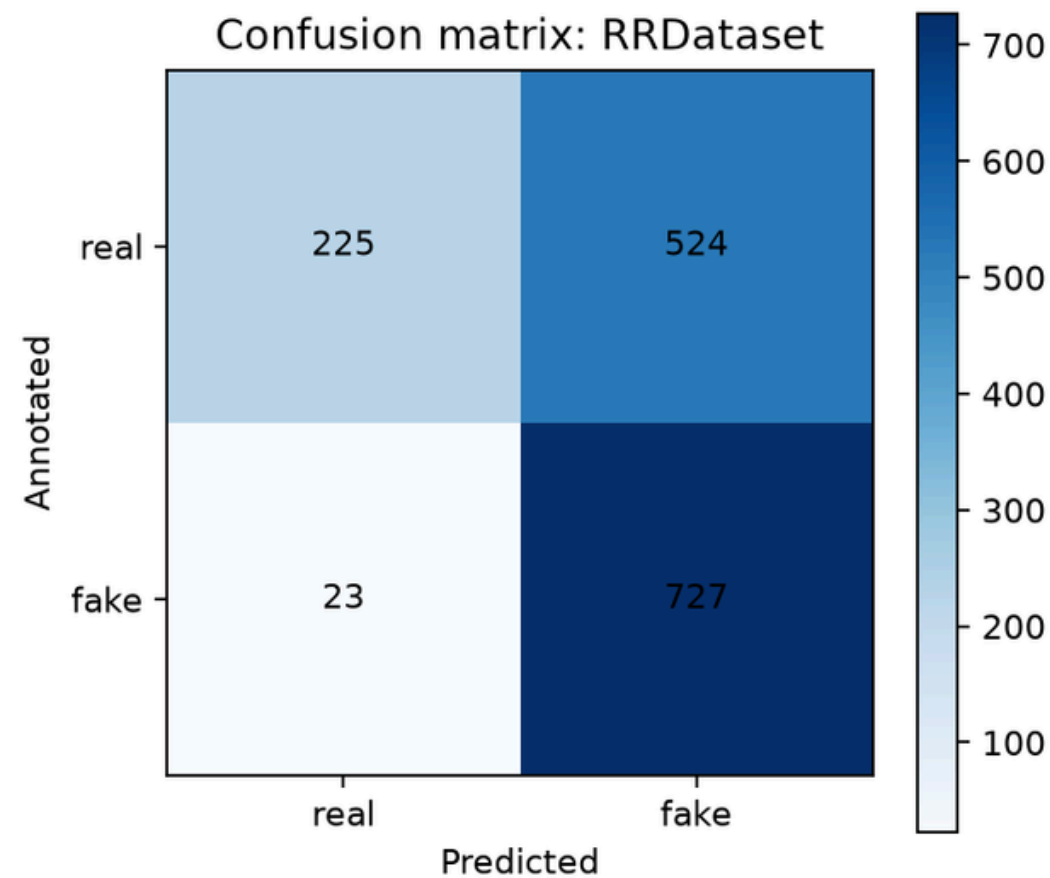
Benchmark by Source Dataset

The balance between missed fakes and false alarms varies substantially across source datasets.

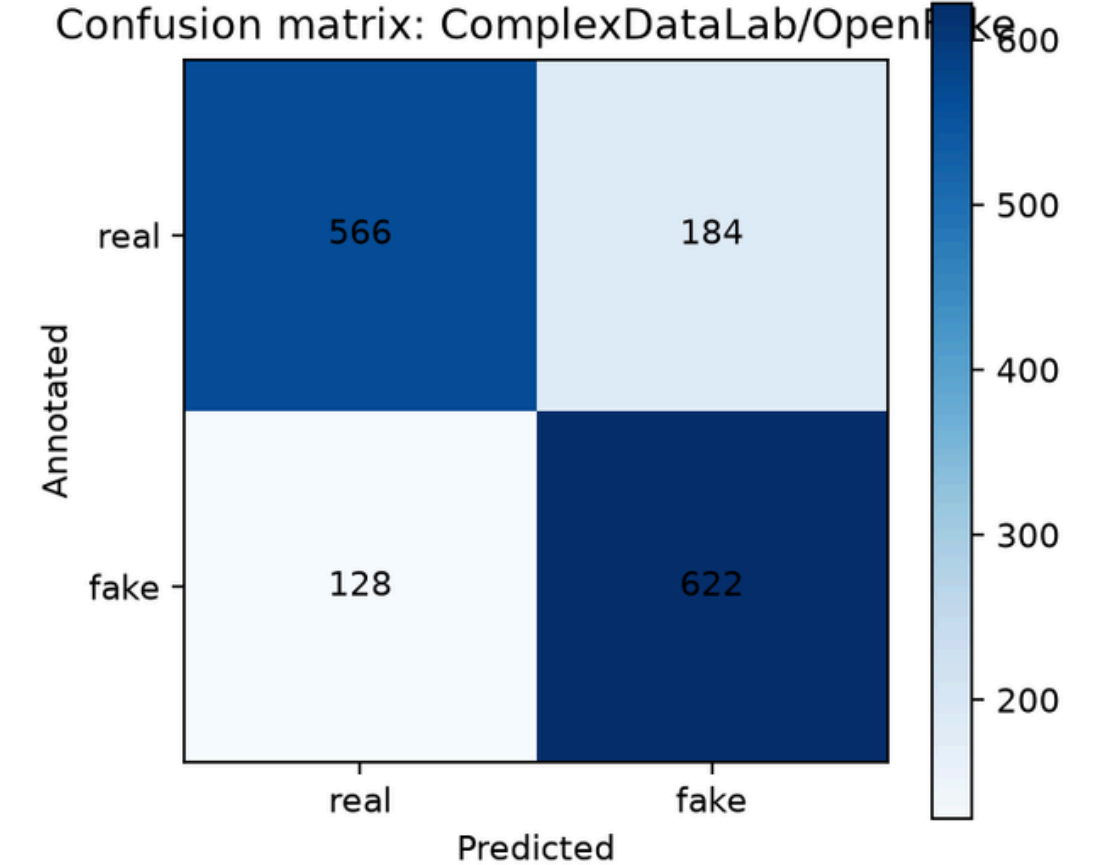
SID-Set



RRDataset



OpenFake



Error rate	SID-Set	RRDataset	OpenFake
False-alarm rate Real → Fake	36.7%	70.0%	24.5%
Missed-fake rate Fake → Real	1.6%	3.1%	17.1%

Benchmark Results

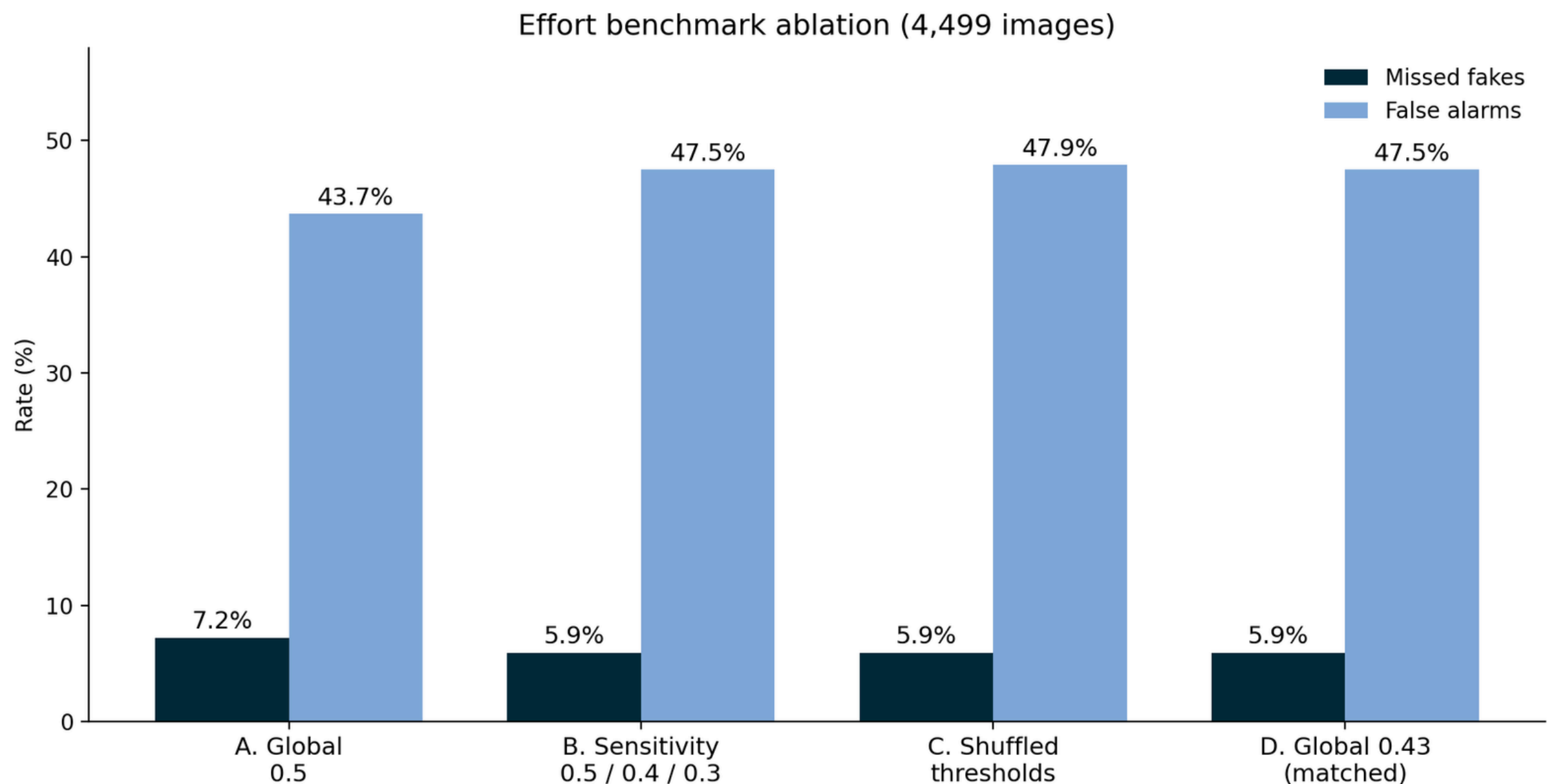
Evaluating EFFORT on 4,499 images across sensitivity levels.

Sensitivity	Accuracy	Fake Recall	Fake F1
Low	0.711	0.91	0.76
Medium	0.75	0.92	0.78
High	0.88	0.97	0.92



Fake-detection performance varies across both source datasets and sensitivity levels. RRDataset shows its strongest F1 on high-sensitivity images, while SID_Set shows a substantial decline for that group. OpenFake is comparatively consistent across sensitivity levels.

What the Benchmark Ablation Tells Us



36% vs 51%

High false alarms: global 0.43 vs 0.3

7.2% → 5.9%

Missed fakes: global 0.5 → sensitivity

5.9%

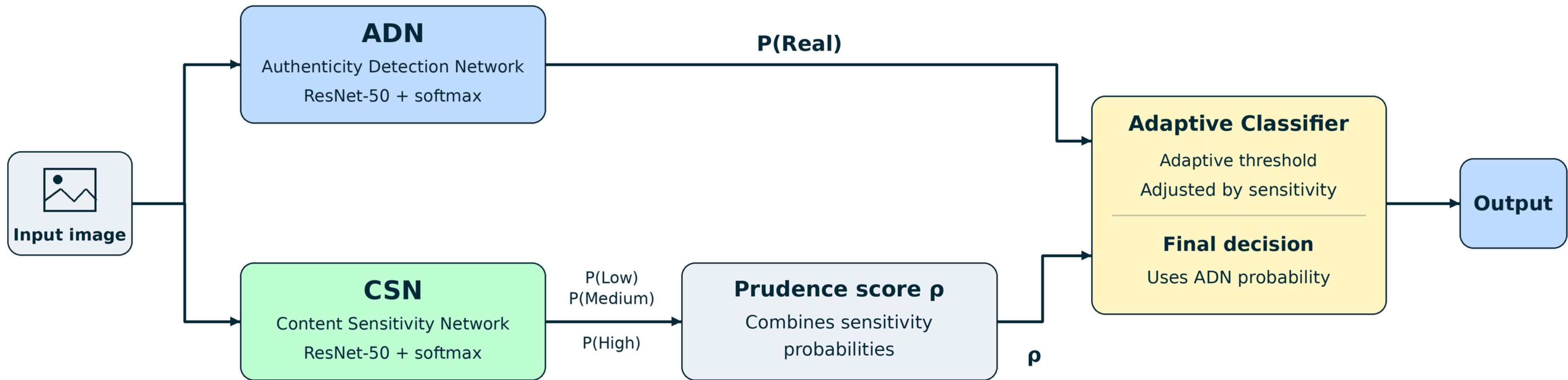
Missed fakes: shuffled thresholds

Four variants

A · Global	B · Sensitivity	C · Shuffled	D · Matched global
One threshold: 0.50	Low 0.50 · Medium 0.40 High 0.30	B's thresholds randomly reassigned across images	One threshold: 0.43 Matches B's overall false-alarm rate

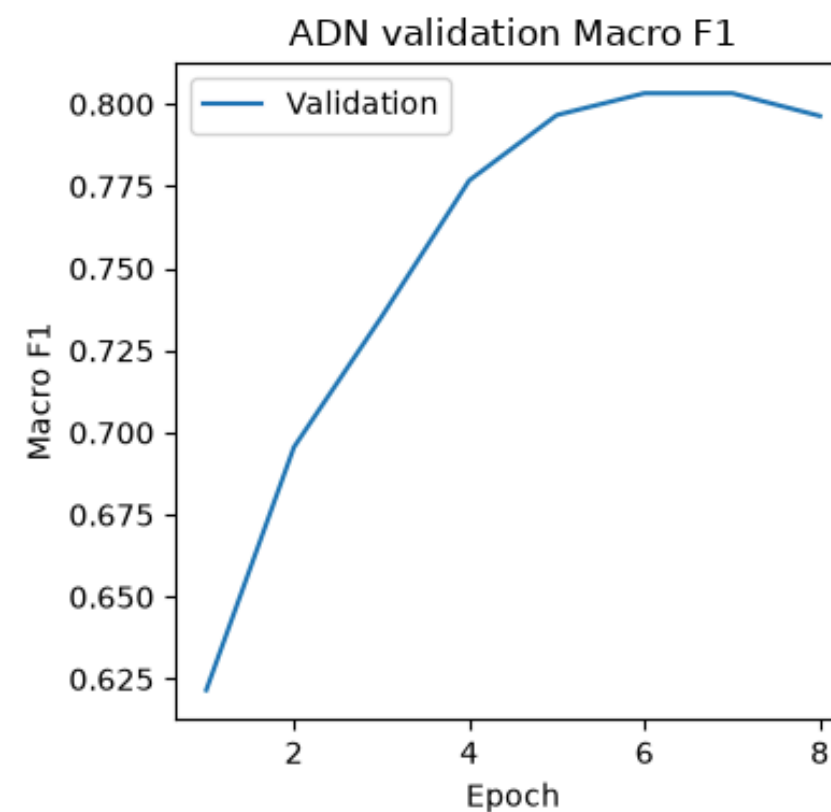
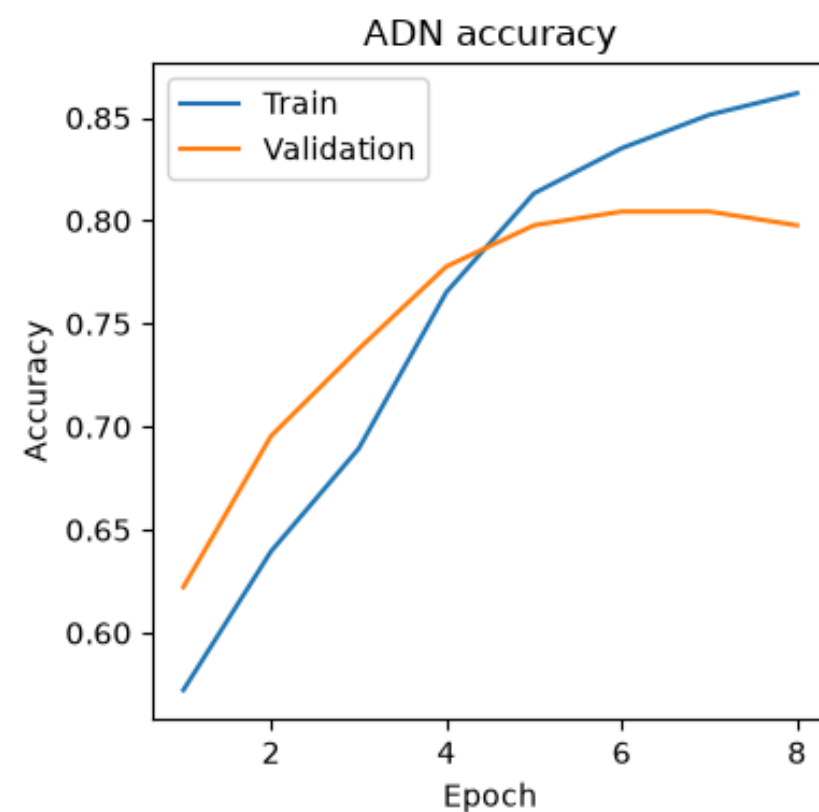
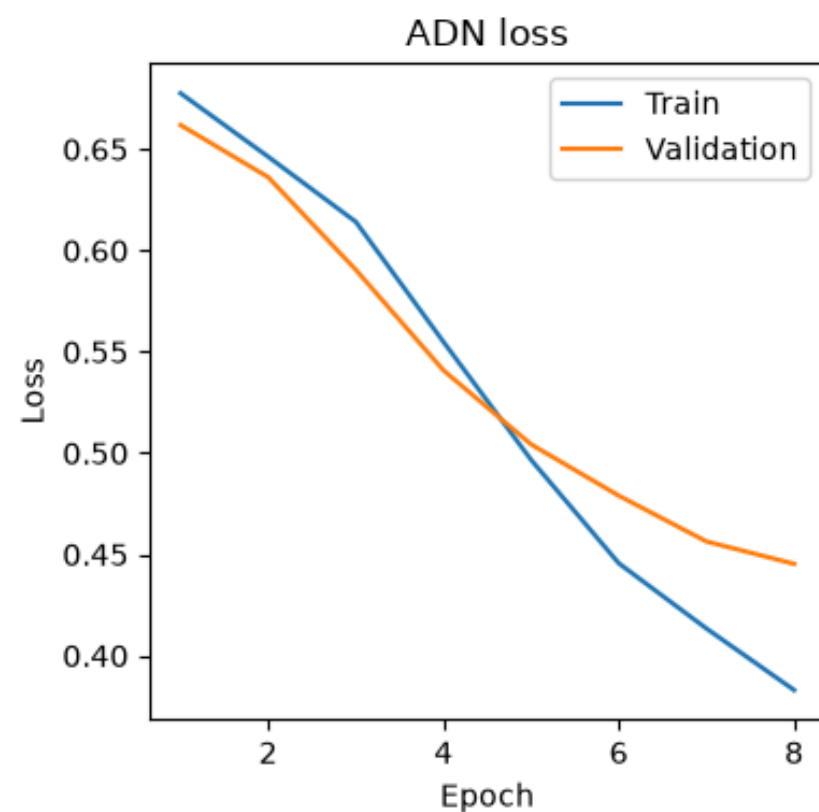
Sensitivity-Aware Deepfake Detection

Two networks analyse the same image; sensitivity adjusts the final decision threshold.

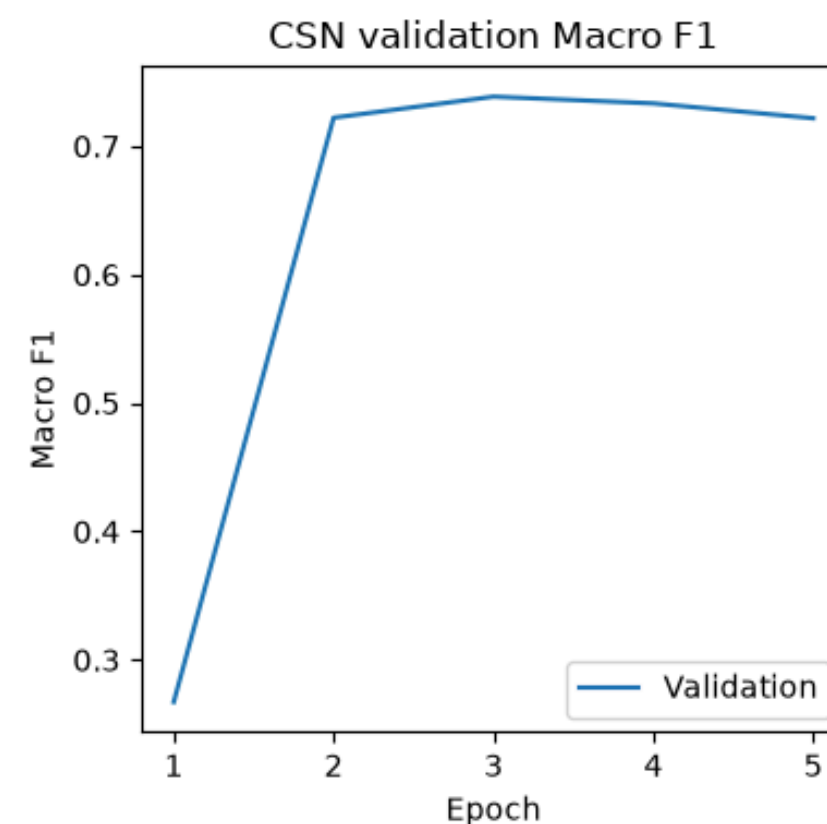
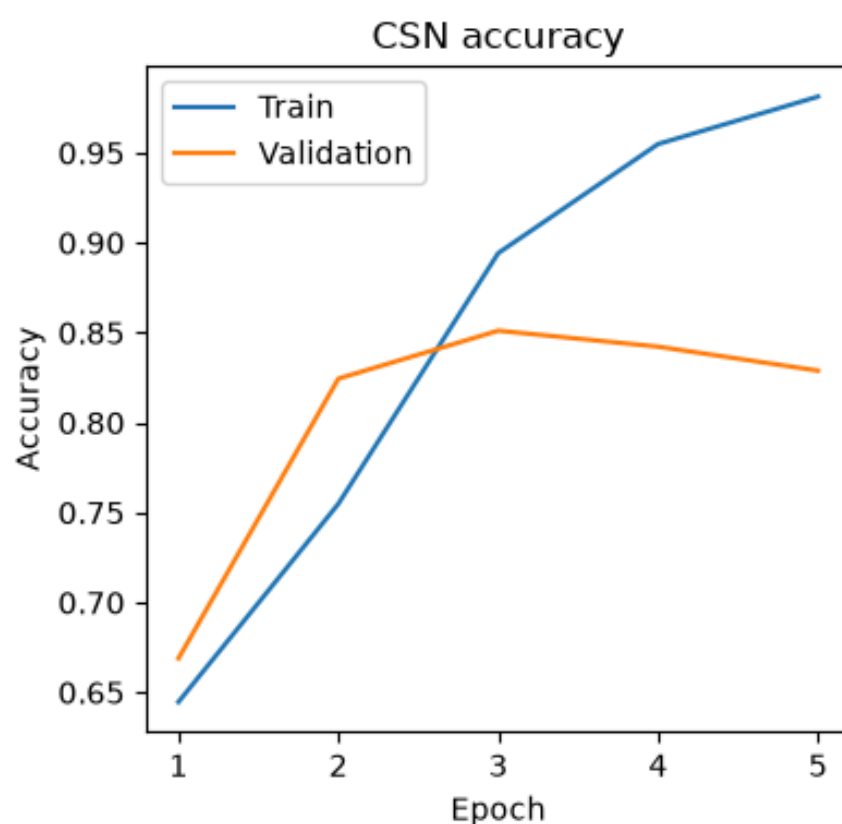
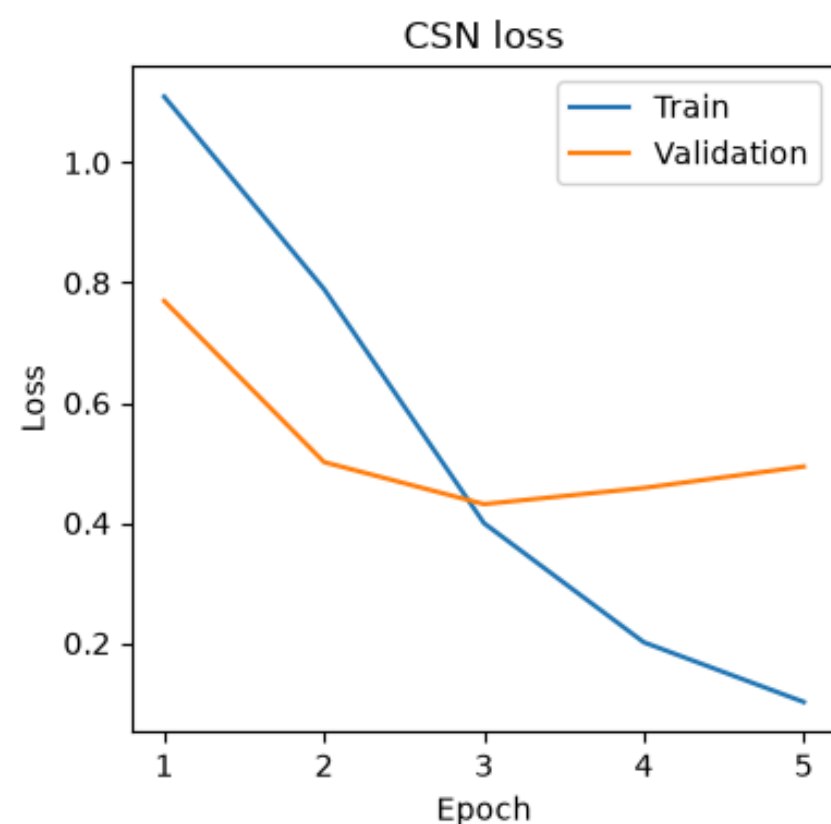


Higher sensitivity → higher threshold → stronger evidence required to classify an image as Real

ADN & CSN — Training Results

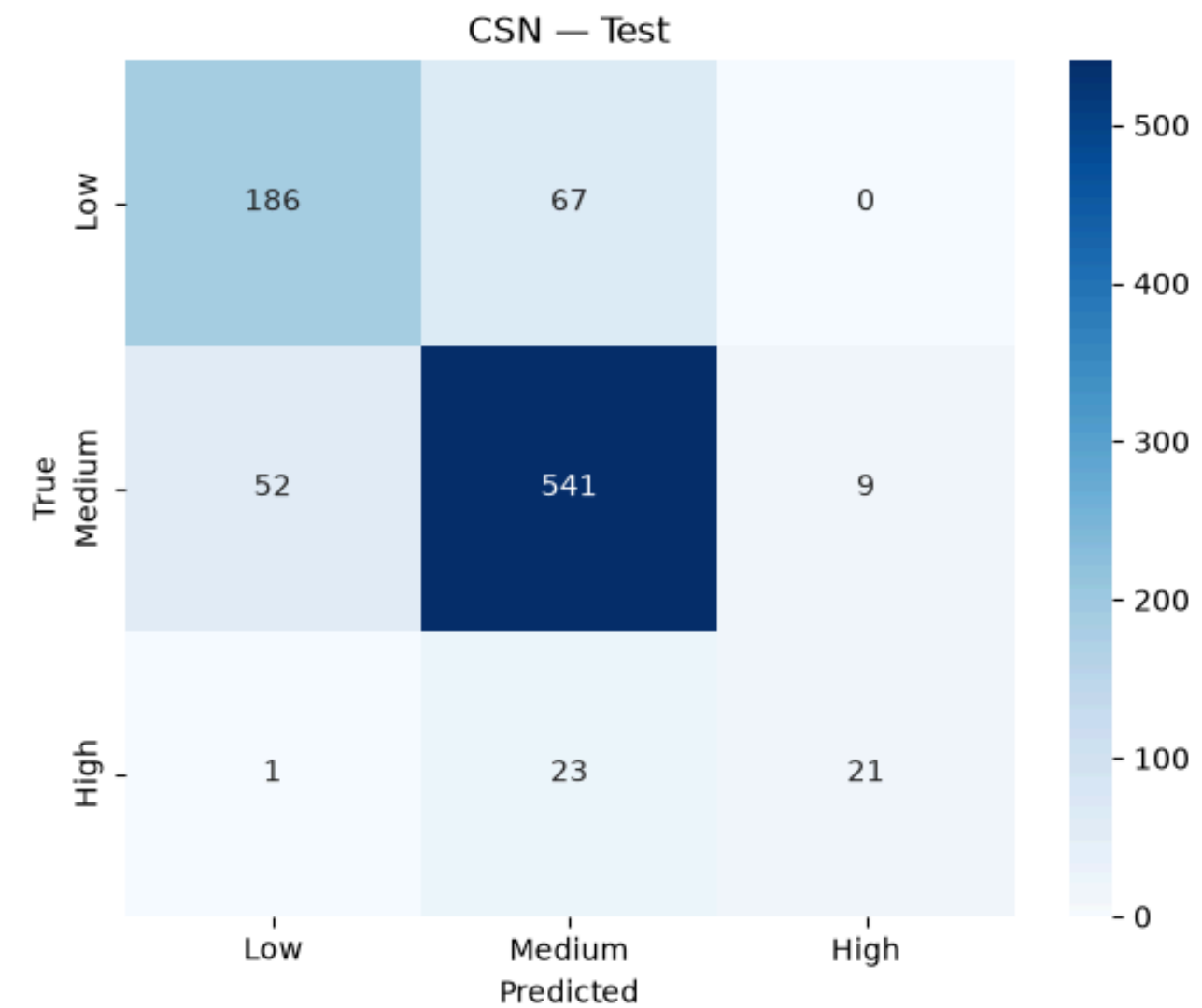
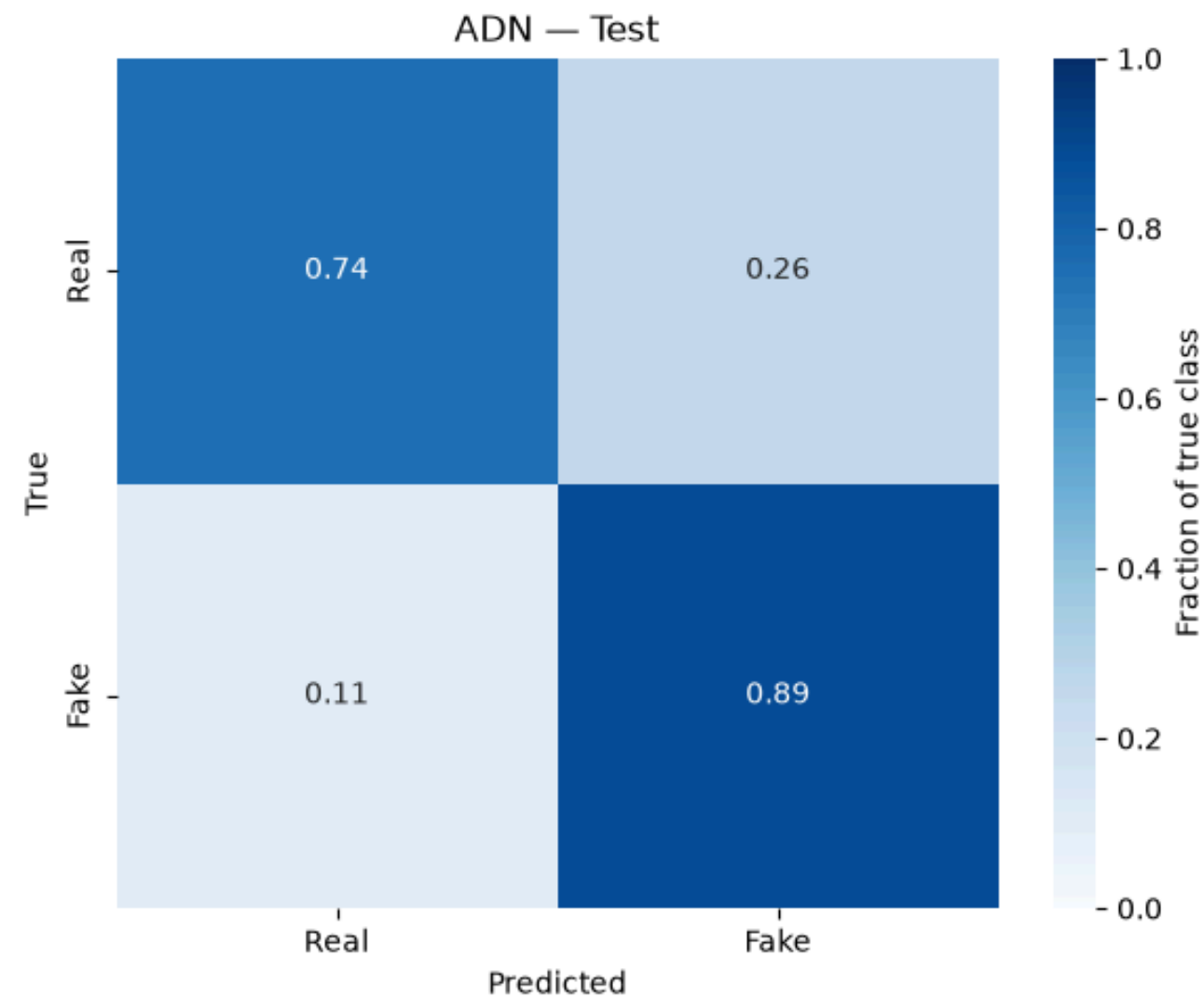


- ADN · Best epoch 6**
- **80.4% Validation accuracy**
 - **0.803 Validation Macro-F1**



- CSN · Best epoch 3**
- **85.1% Validation accuracy**
 - **0.739 Validation Macro-F1**

ADN & CSN — Test Results



ADN	Progressive fine-tuning: classifier first, then the last ResNet block. Dropout (0.3), label smoothing (0.05) and conservative crops.
CSN	High-sensitivity samples: ×2 training weight. Human annotations: ×1.5 training weight.

The Fusion Formula

Two networks analyse the same image; sensitivity adjusts the final decision threshold.

Prudence score

$$\rho(x) = p_L(x) w_L + p_M(x) w_M + p_H(x) w_H$$

p_L, p_M, p_H : CSN probabilities for Low, Medium and High sensitivity

$$w_L = \frac{1}{3} \quad w_M = \frac{2}{3} \quad w_H = 1$$

Adaptive threshold

$$\tau_{\text{Real}}(x) = 0.5 + \varepsilon \rho(x)$$

ε : sensitivity influence

ε is calibrated on validation

Real if $P(\text{Real} | x) > \tau_{\text{Real}}(x)$; Fake otherwise

Higher sensitivity → stronger evidence required to classify an image as Real

How it started!!

weight of the parameters

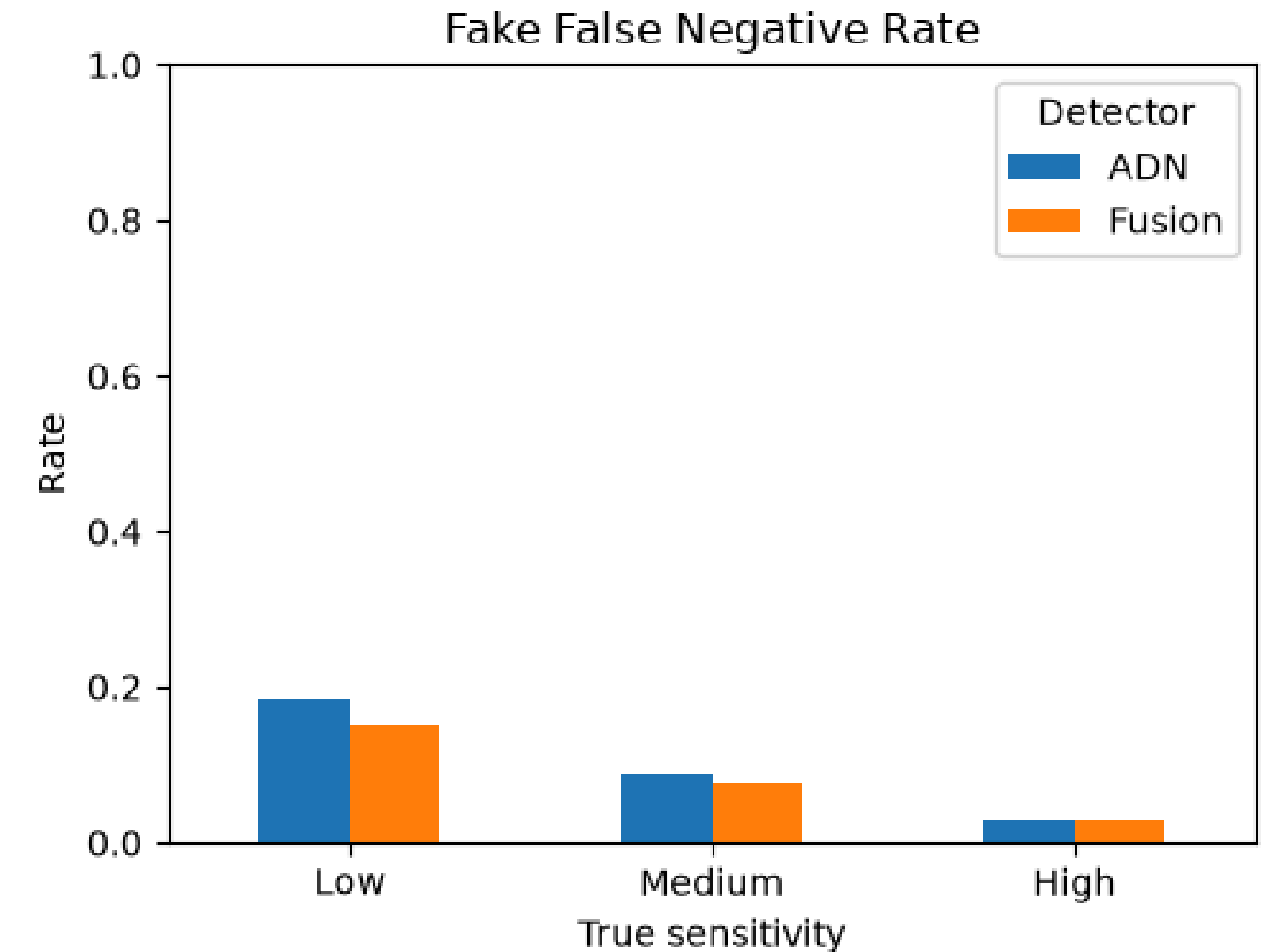
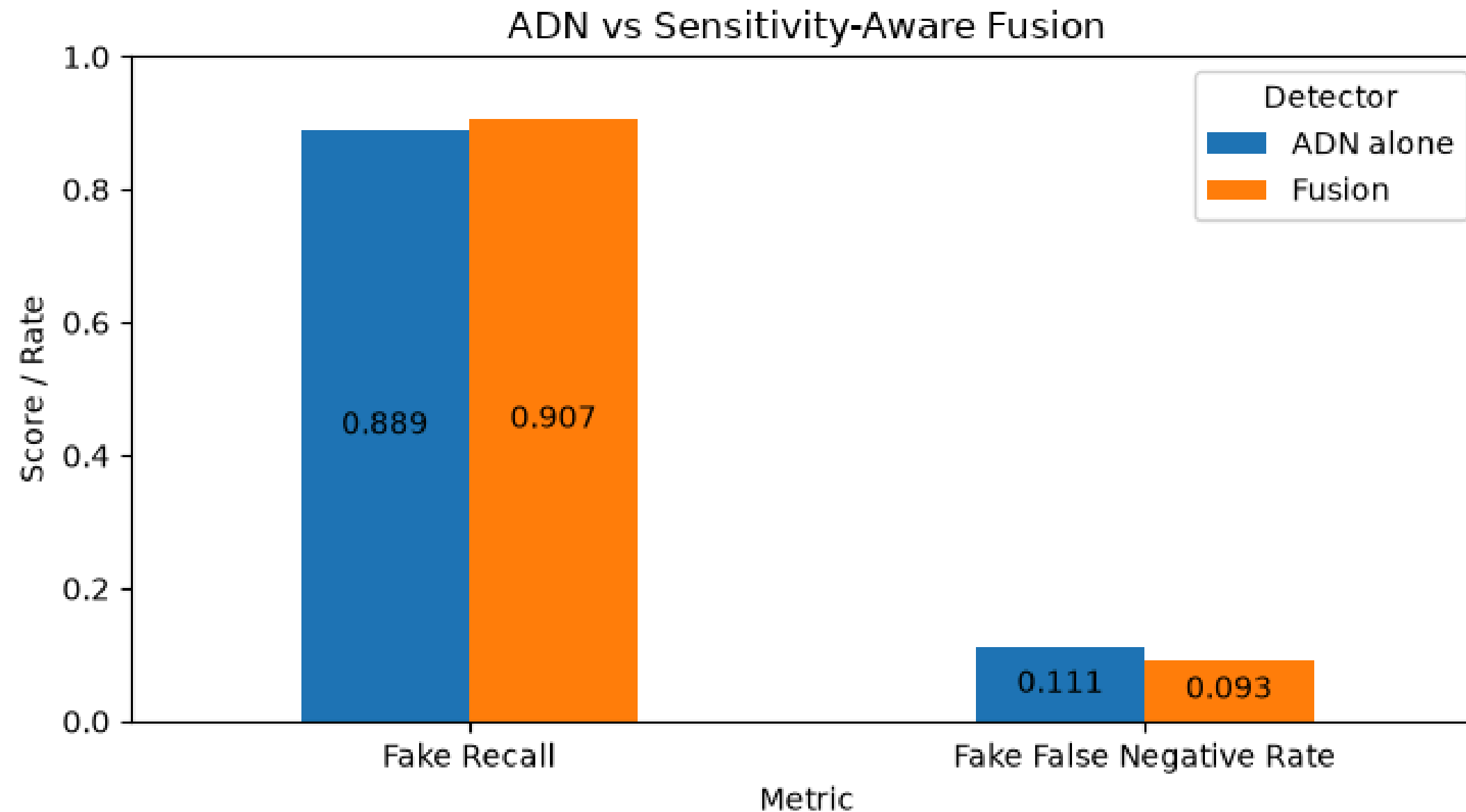
$$\text{Prudence score} = v_0 \alpha + v_1 \beta + v_2 \gamma \quad (\alpha < \beta < \gamma)$$

hyperparameter λ ← values of CSN

$$\text{threshold} = 0.5 + \lambda \text{ prudence score}$$

if $P_{\text{real}} < \text{threshold}$ then fake
if $P_{\text{real}} \geq \text{threshold}$ then real

Final Results

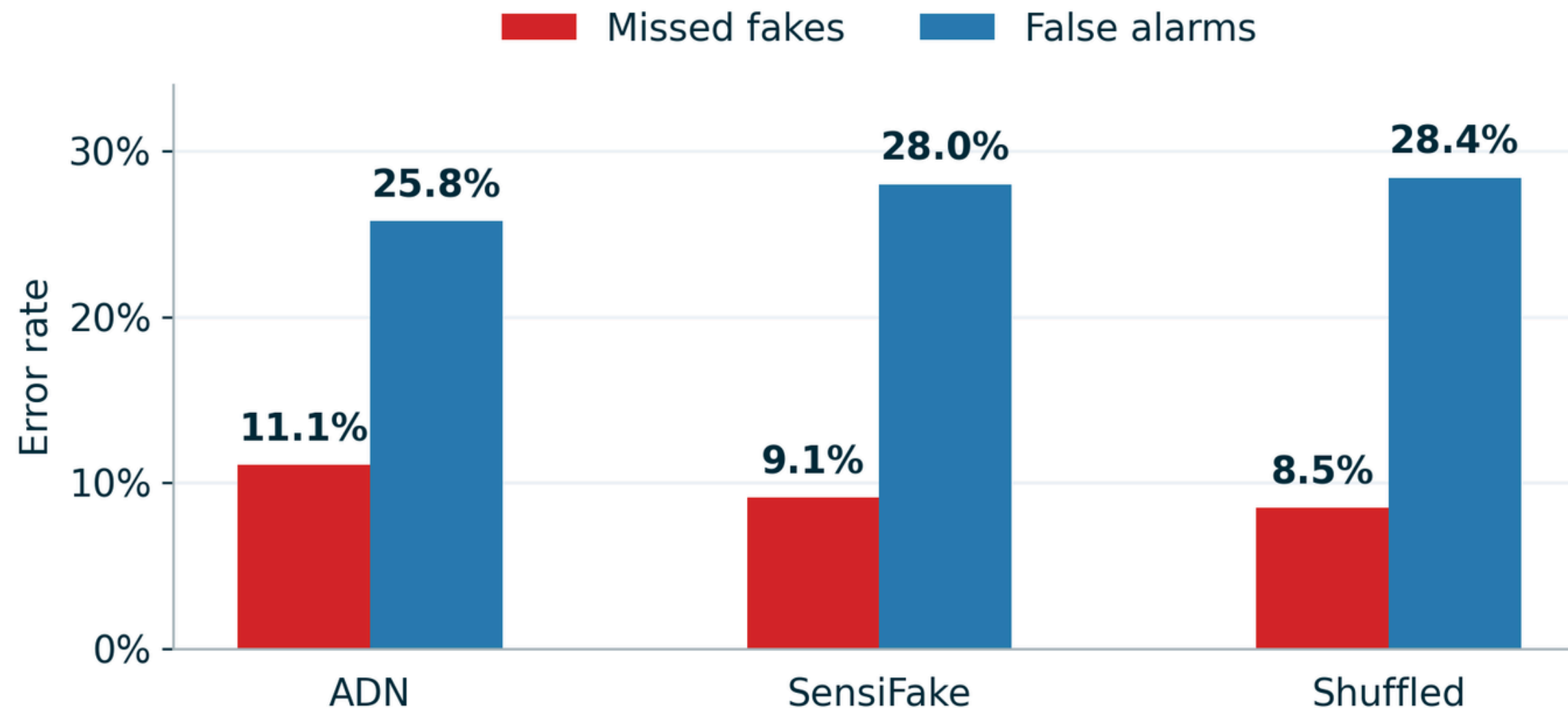


Fusion settings were selected on training/validation data and kept fixed during testing.

- **Fewer missed fakes:** Fake recall increases from **88.9%** to **90.7%**, reducing missed fakes from 50 to 42.
- **Trade-off:** Fusion changes **18 decisions: 8 corrections** and **10 regressions**. Accuracy slightly decreases from **81.6%** to **81.3%**.
- **High-sensitivity limitation:** The mean fusion threshold increases with sensitivity, but High FNR remains unchanged. CSN identifies only **46.7%** of High samples.

Ablation: Is It the Sensitivity?

Does sensitivity-guided thresholding outperform random threshold assignment?



Three variants

- **A • ADN:** fixed threshold, 0.5.
- **B • SensiFake:** sensitivity-dependent threshold.
- **C • Shuffled:** same thresholds, randomly reassigned; mean over 1,000 shuffles.

No clear sensitivity-specific benefit

Shuffled thresholds achieve a similar trade-off. The reduction in missed fakes may mainly reflect a stricter decision threshold.

Conclusions — A More Cautious Detector

A first step towards accounting for the cost of missed fakes.

What we built

A sensitivity-labelled dataset and a detector that adapts its decision threshold to content sensitivity.

What we achieved

50 → 42

missed fakes

10 additional false alarms

Accuracy: 81.6% → 81.3%

16%

fewer missed fakes

8 additional fakes detected

2%

of decisions changed

18 out of 900 test images

−0.22 pp

accuracy

Small overall accuracy decrease

A modest but measurable shift towards caution, at the cost of additional false alarms

Responsible Use & Next Steps

Greater caution comes with responsibility.

Support human review

A fake prediction should prompt verification before an accusation

Improve High-sensitivity recognition

Expand human annotations and strengthen CSN performance.

Validate the added value of sensitivity

Compare against a global threshold under the same false-alarm budget



Sensitive content deserves stronger verification before being accepted as authentic.

Thanks